Sharath Chandrashekhara, Ph.D.

Software Architect | GenAI for Health

San Jose, CA | https://sharathcshekhar.github.io | LinkedIn | Google Scholar

Summary

Experienced **Software Architect** with 10+ years of industrial experience and a **Ph.D. in Computer Science** specializing in data management. Proven track record in translating nascent research challenges into stable, high-availability, production-grade cloud systems. Core domain expertise spans **GenAI systems for personalized wellness**, **HIPAA-compliant digital health platforms**, and **distributed systems and storage**.

Professional Experience

Samsung Research America (Mountain View, CA) **Sep '18 – Present**

Staff Research Engineer II — GenAI & Systems Architecture *Feb '24 – Present*

- Lead systems and infrastructure strategy in Digital Health Lab, architecting modular, event-driven cloud frameworks for LLM-based health and wellness applications across mobile and wearable platforms.
- Oversee LLMOps, prompt management pipelines, benchmarking, guardrails, and chatbot evaluation frameworks.

Staff Research Engineer — Systems & Digital Health Platform *Feb '22 – Feb '24*

- Architected flagship HIPAA-compliant healthcare SaaS platform, driving data ingestion and microservices for clinical trials, graduating prototypes into the commercial Samsung Health Stack product.

Senior Research Engineer — Digital Health Lab *Sep '18 – Feb '22*

- Engineered data collection pipelines and ML-driven biomarker analysis tools deployed across partner hospitals.

Amazon Web Services (AWS) — *SDE Intern, Redshift Database Engines* **Summers '13, '14, '15**

Engineered LZO compression algorithms, table-level backup/restore, and disk failover tooling inside Redshift.

NetApp (Bangalore) & **NDS / Cisco** (Bangalore & Paris) — *Software Engineer* **Jul '09 – Jul '12**

Developed virtualized enterprise storage controller software (NetApp) and digital TV middleware (NDS/Cisco).

Education

Ph.D. in Computer Science and Engineering — University at Buffalo (SUNY) **Aug '14 – Feb '19**

Dissertation: Flexible Data Management on Mobile Systems (Advisors: Dr. Steven Ko, Dr. Karthik Dantu).

M.S. in Computer Science — University at Buffalo (CGPA: 3.84/4.00) **Aug '12 – Sep '14**

B.E. in Electrical and Electronics Engineering — SJCE, Mysore (First Class with Distinction) **Oct '05 – Jul '09**

Selected Publications & Patents

- **MAD-Fusion**: Modality-Aware Dynamic Fusion in Activity Recognition. *IEEE Internet of Things Journal 2026.*
- **MindfulBuddy**: Breathing Biomarkers from Earbud Motion Sensors. *IEEE Internet of Things Journal 2025.*
- **Cough-DL**: Deep Learning Model for Ear-Worn Cough Detection. *IEEE EMBC 2024.*
- **BlueMountain**: Customized Data Management on Mobile Systems. *ACM MobiCom 2017 & HotStorage '15.*
- **Cider**: Block-Level Variable Redundancy on Flash Arrays. *IEEE ICCCN 2017 & US Patent WO2015161140A1.*